Notes: Kremer (QJE, 1993)

The O-Ring Theory of Economic Development

Contents

1 Big picture

- **Question.** Why are productivity, wages, firm size, technology choice, and educational outcomes so different across countries and firms, even when measured differences in individual skills may be modest?
- **Core idea.** Many production processes are like an O-ring: output can be sharply reduced by a mistake in any one of many tasks. In such production, worker qualities are complements.
- **Production mechanism.** Output is the product of success probabilities across tasks. If every task has to go right, small differences in average task reliability compound into large differences in output.
- **Matching result.** Workers of similar skill optimally match with each other. High-quality workers are most valuable when teamed with other high-quality workers, so equilibrium has positive assortative matching.
- **Wage implication.** Wages rise steeply in skill because a slightly better worker raises the expected value of the entire production chain.
- **Development implication.** Countries with higher average skill can have much higher output, larger firms, more complex technologies, and less agriculture, even if underlying skill gaps look small.
- **Firm implication.** Within firms, wages of workers in different occupations should be positively correlated because firms sort by quality across tasks.
- **Human-capital implication.** If skill is imperfectly observed, matching is imperfect. This creates spillovers and strategic complementarities in education: one worker's incentive to invest depends on the expected quality of coworkers.
- **Multiple-equilibrium implication.** Under imperfect observability, countries or groups can be stuck in low-education, low-skill equilibria even when a high-education equilibrium also exists.
- **Policy implication.** Education subsidies can be welfare-improving and can have multiplier effects because they raise both own skill and the expected skill of future coworkers.
- **Takeaway.** The paper offers a microeconomic theory of complementarities in worker quality that connects matching, wages, technology complexity, development, and human-capital externalities.

2 Incremental contribution of this paper

2.1 Contribution relative to standard human-capital and development models

- Standard human-capital models often treat labor quality as efficiency units. Under that view, several low-skill workers can substitute for one high-skill worker.
- Kremer introduces a production structure in which quality and quantity are not freely substitutable. A bad component or bad task can sharply reduce the value of the whole product.
- This change in the production function generates much larger wage and productivity differences from small quality differences.

Incremental contribution: The paper's key incremental contribution is to replace additive or efficiency-unit labor aggregation with a multiplicative task-success technology. That single modeling change produces assortative matching, steep wage schedules, technology sorting, and large development gaps.

2.2 Contribution relative to Rosen, Lucas, and hierarchy models

- Rosen-style superstar models and Lucas-style span-of-control models emphasize the multiplicative importance of managerial quality.

- Kremer shifts attention from vertical hierarchy to horizontal matching: workers at the same level of production interact through task complementarities.
- High-skill workers do not merely supervise more people; they work with other high-skill coworkers in high-quality production chains.

Incremental contribution: The paper extends the idea of multiplicative quality effects from managers and superstars to teams of ordinary tasks. This helps explain firm-level sorting and the correlation of wages across occupations inside firms.

2.3 Contribution relative to Becker-style matching

- Becker's matching logic explains positive assortative matching when complementarities are present.
- Kremer provides a concrete production technology that generates those complementarities.
- The positive cross-partial of output in worker skills gives a direct production-based reason for positive assortative matching.

Incremental contribution: The paper imports matching theory into production and development: workers are not just paid their standalone productivity; their value depends on the quality of teammates.

2.4 Contribution to human-capital externalities

- In the perfect-information benchmark, matching is efficient and education choices are privately optimal.
- With imperfectly observed skill, workers are not perfectly matched, and each worker's education raises the expected quality of his future coworkers.
- This generates positive spillovers, strategic complementarities, and possible multiple equilibria.

Incremental contribution: The paper provides a microfoundation for human-capital externalities that does not require direct knowledge spillovers. The externality comes from labor-market matching under imperfect information.

2.5 Contribution to development economics

- The paper links micro production complementarities to macro development facts: cross-country income gaps, small firms in poor countries, simple technologies in low-skill economies, and agricultural shares that fall with development.
- It also provides a mechanism for low-level traps: low expected coworker quality lowers education incentives, which validates low expected coworker quality.

Incremental contribution: Kremer's incremental contribution is to show that development gaps can be amplified by complementarities in quality, not only by capital scarcity, technology gaps, or aggregate increasing returns.

3 Model primitives and notation

3.1 Tasks and worker quality

- Production consists of n tasks.
- Task i is performed by one worker with quality q_i.

- Quality q_i is the expected fraction of maximum product value retained if worker i performs the task.
- Quality may reflect the probability of doing the task perfectly, or the expected severity of mistakes.
- Mistakes by different workers are independent.

Interpretation: A worker's skill is not merely how many units of work he can produce. It is the reliability with which a critical task is completed without damaging the whole product.

3.2 Capital and maximum output

- Capital enters in Cobb-Douglas form.
- B denotes output per worker with one unit of capital if all tasks are performed perfectly.
- The economy has a fixed supply of capital k^*.
- Workers supply labor inelastically.

Interpretation: Capital is conventional; the distinctive feature is the multiplicative interaction among worker qualities.

3.3 O-ring production function

Expected output is:

$$E(y) = k^{\alpha} \left(\prod_{i=1}^{n} q_i \right) nB. \tag{1}$$

- If all $q_i = 1$, the firm obtains full output $k^{\alpha} nB$.
- If one q_i is low, total output falls proportionally with that task's reliability.
- The product form means quality mistakes compound across tasks.

Interpretation: This production function formalizes the Challenger/O-ring example: many components must work together, and a failure in one component can destroy the value of the whole project.

3.4 Difference from efficiency units

- In an efficiency-units model, two workers with half the skill can often substitute for one worker with full skill.
- In the O-ring model, two mediocre workers cannot generally substitute for one excellent worker in a critical task.
- Quantity cannot freely replace quality inside a given production chain.

Interpretation: This is why the O-ring model generates much sharper sorting and wage patterns than standard labor aggregation.

4 The O-ring equilibrium: matching and wages

4.1 Firm problem

A firm facing wage schedule $w(q)$ and rental rate r chooses capital k and worker qualities q_i to maximize:

$$\max_{k,\{q_i\}} k^\alpha \left(\prod_{i=1}^{n} q_i\right) nB - \sum_{i=1}^{n} w(q_i) - rk. \tag{2}$$

The first-order condition for worker quality is:

$$\frac{dw(q_i)}{dq_i} = \frac{dy}{dq_i} = \left(\prod_{j\neq i} q_j\right) nBk^\alpha. \tag{3}$$

Interpretation: The marginal value of a worker's skill depends on the quality of all the other workers. A worker is more valuable in a high-quality team.

4.2 Positive cross-partials and assortative matching

The cross derivative of output in the qualities of two different workers is positive:

$$\frac{d^2 y}{dq_i d\left(\prod_{j\neq i} q_j\right)} = nBk^\alpha > 0. \tag{4}$$

- A high-quality worker raises output more when other workers are high quality.
- A low-quality worker causes more damage in a high-quality production chain.
- Therefore, firms optimally match workers of similar quality together.

Interpretation: This is the central assortative matching result. High-q workers sort into high-q firms, and low-q workers sort into low-q firms.

4.3 Wage schedule under perfect matching

Under perfect matching, all workers in a firm have the same q. The first-order condition becomes:

$$\frac{dw(q)}{dq} = q^{n-1} nBk^\alpha. \tag{5}$$

With the capital first-order condition substituted in, the integrated wage schedule is:

$$w(q) = (1-\alpha)(q^n B)^{1/(1-\alpha)} \left(\frac{\alpha}{r}\right)^{\alpha/(1-\alpha)} + c, \tag{6}$$

or equivalently,

$$w(q) = (1-\alpha)q^n Bk^\alpha + c. \tag{7}$$

Interpretation: Wages rise steeply in q because q enters as q^n. The more tasks in production, the stronger the effect of skill on wages.

4.4 Competitive equilibrium

- Firms are indifferent across skill levels once wages adjust to productivity.
- Workers of the same skill are matched together.
- The constant of integration c adjusts so the capital market clears.
- Since firms have zero profits, equilibrium is efficient under perfect matching.

Interpretation: Without information problems, the model does not generate inefficient underinvestment by itself. Inefficiency enters through imperfect observability and imperfect matching.

5 Applications to development and labor markets

5.1 Large cross-country wage and productivity gaps

- Small differences in worker reliability can generate large differences in output because reliability compounds over tasks.
- If production has many tasks, the output ratio between high-skill and low-skill teams can be very large even when each individual skill difference is modest.
- This helps explain why measured productivity differences across countries can be enormous.

Interpretation: The O-ring production function amplifies small skill gaps into large income gaps. It therefore provides a mechanism for large cross-country differences without requiring equally large differences in individual ability.

5.2 Positive correlation of wages within firms

- Because high-quality workers match together, firms with high-quality accountants also hire high-quality engineers, managers, and production workers.
- This implies a positive correlation between wages of workers in different occupations inside the same enterprise.
- The prediction is not only that some firms pay more; it is that pay is systematically high across occupations in high-quality firms.

Interpretation: The model predicts firm-level wage premia arising from worker sorting, not only from rent sharing or internal labor markets.

5.3 Sequential production and agriculture

- In a sequential variant, earlier tasks determine whether later tasks are worth doing.
- High-skill workers should be allocated to later stages where product value is already high.
- Low-skill workers are allocated to earlier stages, including primary production.
- Poor countries therefore tend to have larger primary/agricultural sectors.

Interpretation: This mechanism links development to structural transformation. As skill rises, economies can move into later-stage, high-value production.

5.4 Complex technologies and firm size

When firms can choose the number of tasks, they solve:

$$\max_{n,q} q^n nB(n) - nw(q). \tag{8}$$

The first-order condition for technology choice implies:

$$-\log(q) = \frac{B'(n)}{B(n)}. \tag{9}$$

- Since the left-hand side declines with q, higher-skill workers use higher-n technologies.
- More complex technologies are more valuable when mistakes are unlikely.
- This can explain why rich countries use more complex production methods.

Interpretation: High-skill economies do not merely use the same technologies better. They endogenously choose more complex technologies that would be too fragile in low-skill environments.

6 Human capital, imperfect observability, and multiple equilibria

6.1 Why imperfect matching matters

- With perfect matching, private education choices are efficient.
- With imperfect observability, firms cannot perfectly sort workers by true skill.
- Each worker's expected coworker quality depends partly on the education choices of others.
- This creates externalities in human-capital investment.

Interpretation: The key externality is not a direct knowledge spillover. It is a matching externality: better-educated workers improve the expected quality of the teams available to others.

6.2 Stochastic education technology

The education technology is:

$$\log(q) = \log[g(e)] + \varepsilon, \qquad \varepsilon \sim N(0, \sigma_\varepsilon^2). \tag{10}$$

The function $g(e)$ satisfies:

$$g' > 0, \qquad g'' < 0, \qquad g'(0) = \infty, \qquad g(e) > 0, \qquad g(\infty) = 1. \tag{11}$$

Interpretation: Education raises expected skill, but with diminishing returns and idiosyncratic noise.

6.3 Noisy testing and signal extraction

Skill is observed through a noisy test score:

$$\log(t) = \log(q) + \mu, \qquad \mu \sim N(0, \sigma_\mu^2), \qquad cov(\mu, \varepsilon) = 0. \tag{12}$$

The share of test-score variance due to true ability is:

$$\theta = \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_\mu^2}. \tag{13}$$

Given the distribution of scores, firms infer expected skill conditional on the test score and the average skill level chosen by the population.

Interpretation: If tests are precise, firms match close to true skill. If tests are noisy, the average skill of the population becomes more important for each worker's expected productivity.

6.4 Wage under imperfect observability

The expected wage for a worker with test score t when the population skill level is $\bar{q}$ is:

$$w(t, \bar{q}) = t^{n\theta} \bar{q}^{n(1-\theta)} A^n. \tag{14}$$

- Own test score matters with exponent $n\theta$.
- Average population skill matters with exponent $n(1-\theta)$.
- The noisier the test, the more wages depend on population average skill.

Interpretation: Under noisy testing, a worker's payoff depends on the education choices of others. This creates strategic complementarity.

6.5 Strategic complementarity in education

The payoff from choosing education e when others choose $\bar{e}$ is wage minus education cost:

$$V(e, \bar{e}) = [g(e) \exp(\varepsilon + \mu)]^{n\theta} g(\bar{e})^{n(1-\theta)} A^n - e. \tag{15}$$

The cross-partial is positive:

$$V_{12} > 0. \tag{16}$$

- If others invest more in education, expected coworker quality rises.
- This raises the marginal return to own education.
- Education choices are strategic complements.

Interpretation: This is the key mechanism behind multiplier effects and possible multiple equilibria.

6.6 Multiple equilibria and education subsidies

Symmetric Nash equilibria occur where each agent's optimal education equals the education chosen by all others. Multiple equilibria can arise if the reaction function is steep enough.

The slope condition is written in the paper as:

$$\rho = \frac{g'(e)^2 n(1-\theta)}{(1-n\theta) g'(e)^2 - g(e) g''(e)} > 1. \tag{17}$$

- A low-education equilibrium can persist because low expected coworker quality makes education less attractive.
- A high-education equilibrium can also exist because high expected coworker quality raises the return to education.
- Education subsidies can move the economy toward the high equilibrium.
- Small exogenous differences can have large effects through the education multiplier.

Interpretation: The model gives a formal logic for development traps, educational externalities, and self-fulfilling differences between groups or countries.

7 Generalizations and extensions

7.1 General symmetric production functions

The matching result generalizes to symmetric production functions with positive cross derivatives in worker skill. For example:

$$y = \left(\prod_{i=1}^{n} q_i\right)^{\psi}, \qquad 0 < \psi < \frac{1}{n}. \tag{18}$$

Interpretation: Positive assortative matching does not require the exact O-ring form. It requires complementarity between worker qualities.

7.2 Decreasing returns in skill

- With decreasing returns to the skill of the workforce as a whole, matching remains positive if cross-partials are positive.
- However, wage and output differences become less amplified.
- Human-capital accumulation becomes globally concave, so multiple equilibria disappear.

Interpretation: O-ring production is one strong case. The broader lesson is that complementarities determine matching, while the curvature of the wage function determines the possibility of multiple equilibria.

7.3 Negative cross-partials and asymmetric tasks

- Some production functions may have negative cross-derivatives, for example if one worker serves as backup for another in a critical task.
- In such cases, high-skill and low-skill workers may be matched together.
- Asymmetric production functions can allocate workers into occupations depending on which tasks are most sensitive to quality.

Interpretation: The O-ring mechanism is not intended to explain all production. It identifies a class of production processes in which quality complementarities dominate.

8 Tables, figures, and original equation panels

8.1 Panel 1: O-ring production and assortative matching

Read:

- The production function multiplies worker qualities across tasks.
- The firm problem and first-order conditions show that marginal productivity of one worker's skill depends on coworkers' skill.
- The positive cross-derivative implies positive assortative matching.

Economic message: High-skill workers are most valuable when matched with other high-skill workers; low-skill workers impose larger losses in high-quality teams.

value. The probability of mistakes by different workers is indepen-dent. Capital k enters the production function in conventional Cobb-Douglas form and is not differentiated by quality. Define B as output per worker with a single unit of capital if all tasks are performed perfectly. Expected production is thus

$$E(y) = k^{\alpha}(\Pi_{i=1}^{n} q_i)\, nB. \tag{1}$$

Firms are risk-neutral, so the remainder of the paper drops the

Equation (1): O-ring production function

a level of capital, k, and the skill of each worker, q_i, to maximiz revenue minus cost:

$$\max_{k,\{q_i\}} k^{\alpha}\, (\Pi_{i=1}^{n} q_i)\, nB - \sum_{i=1}^{n} w(q_i) - rk. \tag{2}$$

The first-order condition associated with each of the q_i is

$$\frac{dw(q_i)}{dq_i} = \frac{dy}{dq_i} = (\Pi_{j \neq i}\, q_j)\, nBk^{\alpha}. \tag{3}$$

Thus, the increase in output a firm obtains by replacing on worker with a slightly higher skill worker while leaving the skill c its other workers unchanged must equal the increase in its wag bill necessary to pay the higher skill worker. The marginal produc of skill, dy/dq_i, must equal the marginal cost of skill, $dw(q_i)/dq_i$, o else the firm would prefer to employ either lower or higher ski workers.

The search for equilibria can be restricted to those allocation of workers to firms in which all workers employed by any singl firm have the same q. This is because the derivative of the margina product of skill for the ith worker with respect to the skill of th other workers is positive:

$$\frac{d^2y}{dq_i\, d(\Pi_{j \neq i}\, q_j)} = nBk^{\alpha} > 0. \tag{4}$$

Equations (2)–(4): firm problem and positive cross-partial

8.2 Panel 2: Wage schedule and technology choice

Read:

- Equations (5)–(10) derive the equilibrium wage schedule under perfect matching.
- Wages depend on q^n, so skill has a steep effect when there are many tasks.
- Equations (11)–(14) show that higher-skill workers choose more complex technologies with more tasks.

Economic message: The same complementarity that generates assortative matching also explains steep wage schedules and technology sorting.

For now I assume perfect matching; Section II examines imperfect matching.

Given that workers of the same skill are matched together, $q_i = q_i$ for all j and the first-order condition on q can be rewritten as

$$(5) \qquad \frac{dw}{dq} = q^{n-1} nBk^{\alpha}.$$

The first order condition on capital, $\alpha k^{\alpha-1} q^n nB = r$, implies that

$$(6) \qquad k = \left(\frac{\alpha q^n nB}{r}\right)^{1/(1-\alpha)}.$$

It is straightforward to show that payments to capital are αy. The equilibrium rental rate on k, r, will be that which equates the supply of capital, k^*, with the demand, which is given by summing up the capital demanded by the firms hiring all the different skill levels of workers, from zero to one. Since the density of firms hiring workers of a particular skill is $1/n$ times the density of workers of that skill level, this implies that

$$(7) \qquad \int_0^1 \left(\frac{\alpha q^n nB}{r}\right)^{1/(1-\alpha)} \frac{1}{n}\, \partial\phi(q) = k^*.$$

Thus, $r = \alpha B n^{\alpha} [\int_0^1 q^{n/(1-\alpha)} \partial\phi(q)/R^*]^{1-\alpha}$.
(Alternatively, in an open economy, r would be fixed, and k^* would be the equilibrium level of capital.) The first-order condition on q, (5), can be rewritten by substituting in the value of k from equation (6):

$$(8) \qquad \frac{dw}{dq} = nq^{n-1}B\left(\frac{\alpha q^n nB}{r}\right)^{\alpha/(1-\alpha)}.$$

Integrating generates the set of wage schedules that allows firms hiring workers of any single level of skill to satisfy this first-order condition:

$$(9) \qquad w(q) = (1-\alpha)(q^n B)^{1/(1-\alpha)} \left(\frac{\alpha n}{r}\right)^{\alpha/(1-\alpha)} + c,$$

or equivalently,

Equations (5)–(10): wage schedule

$$(11) \qquad \max_{n, \{q_i\}} \left(\Pi_{i=1}^{n} q_i\right) nB(n) - \sum_{i=1}^{n} w(q_i).$$

In equilibrium each firm must satisfy a first-order condition for optimal choice of n and each of the q_i. Since the first-order condition on choice of the q_i is the same as in subsection I.1, the search for equilibria can again be restricted to allocations of workers to firms in which workers of the same skill are matched together, and the firm's problem can be written as

$$(12) \qquad \max_{n,q} q^n nB(n) - nw(q).$$

The first-order condition on choice of n is therefore

$$(13) \qquad q^n B(n) - w(q) + n[\log(q) q^n B(n) + B'(n) q^n] = 0.$$

The first-order condition on q implies that $w(q) = q^n B(n)$, as in subsection I.1. Substituting for $w(q)$ and simplifying,

$$(14) \qquad -\log(q) = B'(n)/B(n).$$

Equations (11)–(14): technology choice

8.3 Panel 3: Imperfect observability and strategic complementarity

Read:

- Education determines expected skill with noise.
- Skill is observed through a noisy test score.
- Firms solve a signal-extraction problem to infer skill from test scores.
- Wages depend on both own signal and average population skill.
- Education choices are strategic complements because higher average education raises the marginal return to own education.

Economic message: Imperfect observability converts a sorting model into a model of human-capital externalities and development traps.

suppose that the production technology is

(15) $$Y = n \,\Pi_{i=1}^{n} q_i,$$

where n is fixed, and there is a stochastic education technology such that skill q depends on e, education or effort,

(16) $$\log(q) = \log[g(e)] + \epsilon \qquad \epsilon \sim N(0,\sigma_\epsilon^2),$$

where

(17) $$g' > 0 \qquad g'' < 0 \qquad g'(0) = \infty \qquad g(e) > 0 \qquad g(\infty) = 1.$$

Skill is observed (even by the worker himself) only through a test score, t, which is a stochastic function of true skill.

(18) $$\log t = \log q + \mu \qquad \mu \sim N(0,\sigma_\mu^2) \qquad \mathrm{cov}(\mu,\epsilon) = 0.$$

The logarithmic form for the errors is chosen so that q takes on only positive values.[10] The error terms ϵ and μ correspond to random

extraction problem to find the conditional expectation of a worker's skill given his test score and the test scores of other agents in the economy.

In equilibrium, all agents choose the effort level $\bar{e}$ and hence the expectation of $\log(q)$ and $\log(t)$ for all agents is $\log(\bar{q})$, where $\bar{q}$ is defined as $q(\bar{e})$. Firms can deduce $\bar{e}$ and thus $\bar{q}$ by observing the distribution of all agents' test scores. Since $\log(q) = \log(\bar{q}) + \epsilon$ and $\log(t) = \log(\bar{q}) + \epsilon + \mu$, and ϵ and μ are independent normals, the conditional distribution of $\log q$ for an agent with test score t given $\bar{q}$ is

(19) $$\log q \,|\, t,\bar{q} \sim N\left(\theta \log t + (1-\theta)\log \bar{q},\ \frac{\sigma_\epsilon^2\sigma_\mu^2}{\sigma_\epsilon^2 + \sigma_\mu^2}\right),$$

where θ is the share of the variance in the test score due to variance in true ability,

(20) $$\theta \equiv \sigma_\epsilon^2/(\sigma_\epsilon^2 + \sigma_\mu^2).$$

Thus, if there is no testing error ($\sigma_\mu^2 = 0$), the expected skill equals the test score, whereas if there is no variation in ability to absorb education ($\sigma_\epsilon^2 = 0$), the expected skill is the average level of skill, $\bar{q}$.

Given the conditional distribution of $\log(q)$ for a single worker

Equations (15)–(18): education and test-score technology

Equations (19)–(20): signal extraction

with test score t, a firm hiring workers of test score t has a conditional distribution of log output of

(21) $$\log\left(n\,\Pi_{i=1}^{n} q_i\right) | t,\bar{q} \sim N\left[\log n + n(\theta \log t + (1-\theta)\log \bar{q}),\ n\frac{\sigma_\epsilon^2\sigma_\mu^2}{\sigma_\epsilon^2 + \sigma_\mu^2}\right].$$

The conditional expectation of output is therefore

(22) $$E\left(n\,\Pi_{i=1}^{n} q_i\right) | t, \bar{q} = n \exp[n(\theta \log t + (1-\theta)\log \bar{q} + \log A)].$$

where A is the constant

(23) $$A \equiv \exp\left[\frac{1}{2}\frac{\sigma_\epsilon^2\sigma_\mu^2}{\sigma_\epsilon^2 + \sigma_\mu^2}\right].$$

This simplifies to

(24) $$E\left(n\,\Pi_{i=1}^{n} q_i\right) | t, \bar{q} = n t^{n}\bar{q}^{n\theta(1-\theta)} A^{n}.$$

By the zero profit condition, the wage is $1/n$ times the expected product:

(25) $$w(t,\bar{q}) = t^{n\theta}\bar{q}^{n(1-\theta)}A^{n}.$$

For $0 < \theta < 1$, each agent's wage is increasing not only in his own test score but also in $\bar{q}$, the skill level chosen by other agents. (Note that in the special case of no measurement error, both θ and A equal one, and the formula for the wage is the same as that derived in Section I under perfect matching.)

The marginal product of education increases with the education of other agents. The payoff V is the wage minus the cost of education:

(26) $$V(e,\bar{e}) = [g(e)\exp(\epsilon + \mu)]^{n\theta} g(\bar{e})^{n(1-\theta)} A^{n} - e.$$

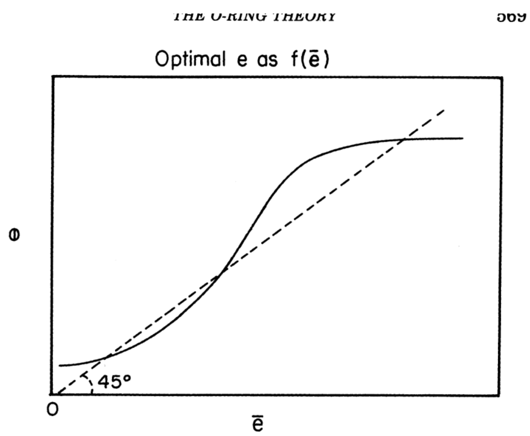
Thus, for any realizations of ϵ and μ, the cross derivative of the payoff with respect to own education and others' education will be positive:

8.4 Panel 4: Multiple equilibria

Read:

- Figure I plots the best response of education to the education level chosen by others.
- Symmetric Nash equilibria occur where the reaction curve crosses the 45-degree line.
- If the slope is greater than one at a symmetric equilibrium, multiple equilibria can arise.

Economic message: Education policy can have multiplier effects because a direct increase in education raises expected coworker quality and thereby induces further education investment.



(a) Figure I: reaction function and possible multiple equilibria.

FIGURE I

Figure I shows the optimal e as a function of the level of $\bar{e}$ chosen by the other agents. Since $g(e) > 0$ and $g'(0) = \infty$, zero education can never be optimal and since $g(e)$ is bounded, the optimal e is bounded. SNE occur where the reaction function crosses the 45 degree line. Cooper and John show that a necessary condition for multiple equilibria is that the slope of the reaction function, ρ, be greater than one at some point, and a sufficient condition is that ρ be greater than one at an SNE. ρ is given by

$$(28)\qquad \rho = -\frac{V_{12}(e,e)}{V_{11}(e,e)} = -\frac{g(e)g'(e)n(1-\theta)g'(\bar{e})}{[(n\theta-1)g'(e)^2 + gb(e)g''(e)]g(\bar{e})}.$$

At an SNE, $e = \bar{e}$, so a sufficient condition for multiple equilibria is that at an SNE,

$$(29)\qquad \rho \equiv \frac{g'(e)^2 n(1-\theta)}{(1-n\theta)g'(e)^2 - gb(e)g''(e)} > 1.$$

For this to hold, the denominator must be positive and smaller in absolute value than the numerator. This implies that

$$(30)\qquad \theta < \frac{g'(e)^2 - g(e)g''(e)}{ng'(e)^2} < 1.$$

(b) Equations (28)–(30): slope condition for multiple equilibria.

8.5 Panel 5: Generalizations

Read:

- Equation (31) illustrates that positive matching can extend to other symmetric production functions.
- The existence of multiple equilibria depends on curvature: decreasing returns weaken the amplification mechanism.

Economic message: The exact O-ring production function is a tractable benchmark; the deeper idea is complementarity in worker quality.

of output in the skill of different workers is positive. Positive cross derivatives could arise for many reasons. For example, doctors, lawyers, and academics often match with similar skill coworkers in hospitals, law firms, and universities. This may be due to learning spillovers within the firm in which high quality workers are better able to teach and learn from their coworkers.

The matching analysis and its implications thus apply to production functions that are homogeneous of degree less than one, such as

$$Y = (\Pi_{i=1}^{n} q_i)^{\psi} \qquad 0 < \psi < \frac{1}{n}. \tag{31}$$

The principal differences under production functions with decreas

9 Short summary

- Kremer introduces an O-ring production function in which output depends multiplicatively on the quality of many tasks.
- The model implies positive assortative matching: high-skill workers work with high-skill workers.
- Wages rise steeply in skill because worker reliability is valuable in proportion to the quality of coworkers and the number of tasks.
- Small differences in average worker quality can generate large differences in output and wages.
- The model explains why rich countries specialize in complex production and large firms, while poor countries have simpler technologies and more primary production.
- If skill is imperfectly observed, matching is imperfect and education creates positive externalities.
- Education choices become strategic complements because each worker's return to education rises with the expected education of coworkers.
- The model can generate multiple equilibria in education and output.
- Education subsidies can improve welfare and have multiplier effects.
- The broader contribution is a production-based theory of complementarities in quality, matching, wages, technology, and development.

10 Conclusion

10.1 Core conclusion

Kremer's O-ring theory shows that when production requires many tasks to be performed well, worker quality is complementary across tasks. This complementarity generates positive assortative matching, steep wage schedules, large productivity differences, and endogenous technology choice. It also creates human-capital spillovers when skill is imperfectly observed.

10.2 Economic intuition

A mistake by one worker can reduce the value of everyone else's effort. Therefore, high-quality workers are most productive with other high-quality workers, and complex technologies are only worthwhile when worker reliability is high. If skill is hard to observe, workers cannot be perfectly sorted; then each worker's education raises the expected quality of future coworkers and increases the return to education for others.

10.3 Incremental contribution

The paper's incremental contribution is to connect four ideas in one unified mechanism: multiplicative production quality, assortative matching, technology complexity, and human-capital externalities. It shows that development traps and cross-country productivity gaps can emerge from micro-level quality complementarities, not only from capital scarcity or aggregate technology differences.

10.4 Takeaway

The O-ring model explains why "quality" matters so much in development. In economies where production requires many interdependent tasks, small improvements in reliability can have large returns, but those returns depend on the quality of the whole team. This creates both powerful gains from coordinated skill upgrading and the possibility of persistent low-skill equilibria.